Mechanical Design and Manufacturing II Final Project

Motor-Powered Rope Traverser

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EXECUTIVE SUMMARY

This project focused on the design and fabrication of a rope-traversing robot powered by a motor-driven four-bar linkage system. The objective was to develop a compact and efficient system capable of moving along a rope while applying mechanical design principles that were learned throughout the ME: 470 course at Michigan State University.

The robot was manufactured using a combination of milling, lathe machining, and 3D printing. Throughout the design process, key challenges were addressed including achieving reliable motion, maintaining balance through the traversal, and modifying the system to meet strict size constraints. Throughout the design phase, several parts of the mechanism were reworked including the lengths of the linkages, the gear typing and ratios, and the footprint of the motor housing.

The final prototype demonstrated performance with minor assistance traversing a 10-meter rope in 78 seconds during Design Day testing. Above all, this outcome was very educational and provided insight into the engineering process which the team members can take when working on future projects. This project taught the effectiveness of mechanical design and iterative problem-solving. Overall, the project demonstrates the integration of mechanical engineering concepts, manufacturing techniques and design optimization in the development of a functional robotic system.

1.0 INTRODUCTION

Mobility systems capable of reliable rope traversal can provide significant advantages while minimizing human effort and risks. These mechanical systems combine transmission elements and structural components to ensure efficient movement across a wire.

In this project, a rope traversing robot capable of autonomously traveling along a horizontal rope was designed and manufactured. To provide motion, linkages such as the ones studied in the ME 470 Design and Manufacturing II course at Michigan State University were incorporated into the design. A transmission system was also developed to supply the robotic arms with enough torque to complete motion. Both kinematic and dynamic analysis were performed on the mechanism to ensure feasibility. The final project was presented during the Design Day competition, which highlighted the knowledge and learning that took place throughout the course.

Our team's goal was to build a mechanism that would cross a ten-meter rope in under one minute. The amount of payload the machine could withstand was also taken into consideration; however, this was not the primary consideration during the design phase. This mechanism was desired to be small enough to fit inside of a standard shoe-box volume of 14" x 9" x 5" at the start of the trial.

Throughout the entire experience, there were several design decisions that had to be re-worked and adjusted to fit the entire project. In the end, the principles of the initial design were sustained but nearly all aspects of the project were changed in some fashion during the design process. This course was very challenging but, in the end, so much learned about the design and manufacturing phases of a project.

2.0 TECHNICAL SUMMARY

To manufacture the best possible rope traversing robot, all group members were tasked with sketching initial design concepts which would traverse the rope in the shortest amount of time. Mechanisms and concepts that were discussed throughout the lectures in ME 470: Design and Manufacturing II at Michigan State University were taken into consideration throughout the design phase.

2.1 Initial Concept Selection

These initial designs took inspiration from the mechanics of previous teams from universities across the United States which completed this project. From there, all group members voted on the configuration with a simple design that was feasible for individuals with little manufacturing experience to produce. A design decision Matrix was formulated from all four initial concept proposals to generate the most realistic design beginners in the manufacturing space could complete. The first proposed idea from Grady suggested the use of two four-bar mechanisms. The initial draft of the mechanism is depicted below in Figure 2.1.a.

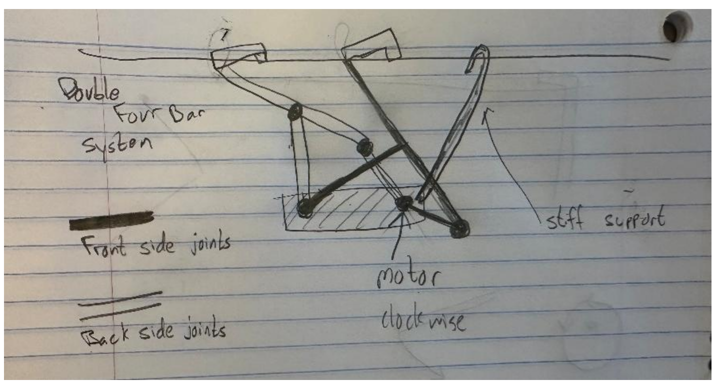

Figure 2.1.a: Initial sketch of a double four bar linkage system proposed by Grady.

The design in Figure 2.1.a displays two different four bar systems that share each other's link one. While the rear support keeps the whole mechanism stable during motion, the four bar systems take turns pulling the body forward. As an opposing idea, Kopiwoda suggested the use of elliptical geared linkages that would be used to drive the motion of the mechanism. His initial design sketch is provided in Figure 2.1.b.

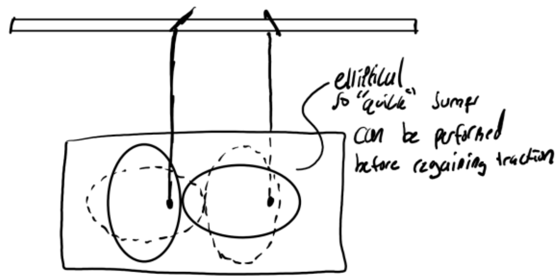

Figure 2.1.b: Gear based jumping robot

The mechanism in Figure 2.1.b is driven by elliptical gears that produce variable velocity motion. This causes the device to move along the rope in periodic rapid "jumps," rather than at a constant speed. During the high-speed phase, the mechanism briefly reduces contact with the rope, effectively creating a short interval of low-friction motion before traction is reestablished. The upper arms then re-engage the rope to restore grip and continue the cycle. The third idea proposed by James revolved around a vertical slider crank-like system. One which the group already had experience in manufacturing in a previous project. That mechanism is shown in design Figure 2.1.c.

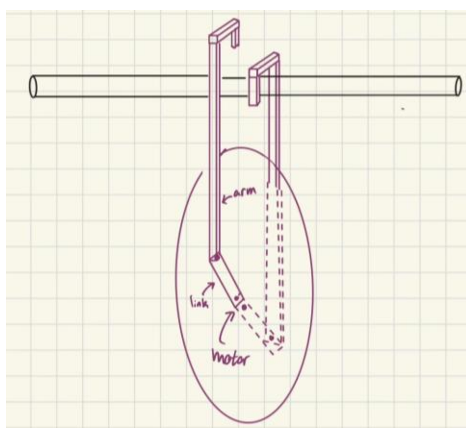

Figure 2.1.c: Egg-shaped crawling robot

The design in Figure 2.1.c is an egg-shaped system using 2 sets of 4-bar mechanisms. Each mechanism moves one of the arms forward in an alternating pattern. This design was not selected because it was unbalanced and would tip every time an arm was lifted and the process of stabilization was too complicated for beginners in the Manufacturing Teaching Laboratory (MTL) to replicate. The fourth and final design that was proposed was generated by Sypitkowski which is modeled below in Figure 2.1.d.

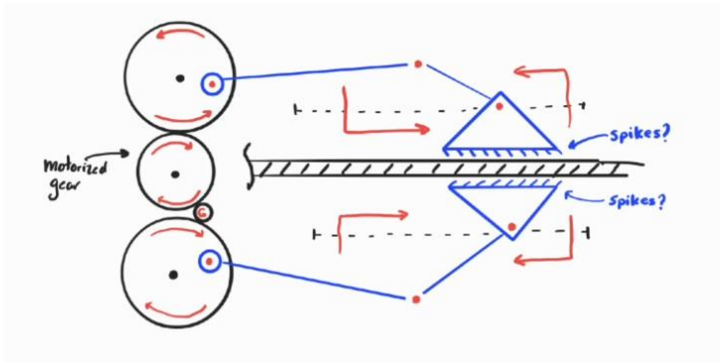

Figure 2.1.d: Double slider-crank spiked machine

The mechanism shown in Figure 4 uses a set of motorized rotating wheels on the left to drive a reciprocating linkage system. As the wheels turn, they pull and push connecting rods (blue links), which transfer motion to the triangular elements on the right. These triangles act like alternating grippers: as one moves forward, it presses spikes or textured surfaces against the central bar, creating traction, while the other retracts and resets. The coordinated back-and-forth motion results in a stepwise linear advancement of mechanism.

Once each of the initial designs were proposed, a decision matrix was generated based on several criteria including ease of manufacturing, design complication, projected speed of movement, total size, and weight. From those projected values the total success of the mechanism was calculated, and the final design was proposed. Below Table 2.1 highlights the decision matrix that was used in the calculations.

Table 2.1: Decision Matrix of the Four Proposed designs pictured above.

	Design A	Design B	Design C	Design D
Manufacturability	4	3	5	3
Complexity	4	2	1	1
Projected Speed	5	1	2	5
Small Size	2	5	4	2
Small Weight	3	1	5	3
Total	**18**	**12**	**17**	**14**

Ultimately from the decision matrix, the four-bar mechanism proposed by Grady was decided on. This idea was then designed using the CAD software SOLIDWORKS as a template of what parts needed to be manufactured. This initial design guided the manufacturing process and helped determine the manufacturing methods that would be used. Figure 2.2 below depicts the CAD assembly of the initial design that was selected to be built.

Figure 2.2: CAD model of initial design previously depicted above in Figure 2.1.a.

From Figure 2.2 above, the design would entail the manufacturing of two four-bar mechanisms and a tail to serve as support. The smallest link attached to the body would be connected to an axel which would rotate to complete the motion of the Grashoff bar mechanism. This mechanism will be discussed further in the future parts of this report.

The selected concept is a motorized lifting mechanism using pivoting link arms and hook supports. The device consists of a fixed base, multiple pivoting link arms, and curved hook interfaces designed to engage with the object being lifted. When actuated, the linkage system rotates upward, raising the supported load.

Compared to the initial concept sketches, the refined design includes a fully defined linkage geometry, mounting base, pivot joints, and attachment points for the actuation system. The mechanism converts rotational input from a motor into controlled angular motion of the arms.

2.2 Four Bar Mechanism

A Grashof condition four-bar mechanism is configured such that at least one link is capable of continuous rotation. When a motor drives link 2, it acts as the input crank, rotating fully about its joint with the fixed body of the mechanism. This rotational motion is transmitted through link 3, known as the coupler which transfers the motion to link 4, the output link. For the case of this project, link 4 was confined to oscillating through a fixed range of motion allowing for the climbing action of the robot to be completed.

The complete actions of the 4-bar mechanism can be modeled using a MATLAB script that was generated throughout the course. From there, the position, velocity, and acceleration of the 4-bar mechanism could be predicted to ensure the motor would be able to successfully power the motion of the mechanism. The so-called kinematic analysis was performed for both the initial proposed design as well as the final design that was presented during design day.

2.2.1 Initial Design Kinematic Analysis

The dynamic analysis shows that the mechanism operates with smooth and predictable motion throughout the input cycle. The position plots indicate a consistent relationship between input and output link angles, confirming proper kinematic behavior. Velocity results reveal periods of higher angular speed in certain links, corresponding to phases where the mechanism delivers greater motion output, while the mechanical advantage plot suggests that force transmission varies over the cycle, peaking when lifting is most effective. Acceleration plots highlight spikes at specific positions, which indicate higher dynamic loads on joints and links during rapid transitions. Overall, the results suggest that the mechanism is kinematically sound, but it experiences non-uniform loading, meaning the motor and components must be designed to handle peak torque and acceleration demands during operation.

A dynamic force analysis was performed for one four-bar side of the climbing mechanism at the instant $\theta_2 = 45°$, with $\omega_2 = 30rpm$ and $\alpha_2 = 0$, to determine joint reactions and the required input torque under the applied rope load. The mass of the mechanism was estimated by modeling each moving link (links 2, 3, and 4) as uniform slender bars with a linear density of $0.5oz/in$, including the additional 6in extension of each coupler. The main body (link 1), shared between the two four-bar systems, was modeled as a lumped mass of 1lb. Based on this model, the total system weight was calculated and equally distributed between the two four-bar linkages, resulting in a constant horizontal resisting force applied at the coupler end of each mechanism.

The input conditions are:

$$\theta_2 = 45°, \quad \omega_2 = 30\text{ rpm} = \pi\text{ rad/s}, \quad \alpha_2 = 0$$

The rope applies a constant horizontal resisting force on the extended coupler equal to half of the total system weight:

$$\overrightarrow{F_H} = \langle -4.935, 0\rangle\text{ N}$$

Link lengths used for kinematics:

$$a = 0.0381\text{ m}, \quad b = 0.1524\text{ m}, \quad c = 0.1524\text{ m}, \quad d = 0.1778\text{ m}$$

Masses:

$$m_2 = 0.02126\text{ kg}, \quad m_3 = 0.17010\text{ kg}, \quad m_4 = 0.08505\text{ kg}$$

Moments of inertia (slender rods):

$$I_{G2} = \frac{1}{12}m_2L_2^2, \quad I_{G3} = \frac{1}{12}m_3L_3^2, \quad I_{G4} = \frac{1}{12}m_4L_4^2$$

From position, velocity, and acceleration analysis:

$$\theta_3 \approx 49.69^\circ, \quad \theta_4 \approx 110.06^\circ$$

$$\omega_3 \approx -0.8193\ \mathrm{rad/s}, \quad \omega_4 \approx -0.07390\ \mathrm{rad/s}$$

$$\alpha_3 \approx 1.5726\ \mathrm{rad/s^2}, \quad \alpha_4 \approx 3.5983\ \mathrm{rad/s^2}$$

Center of Mass Accelerations

$$\overrightarrow{a_{G2}} \approx \langle -0.133, -0.133 \rangle\ \mathrm{m/s^2}$$

$$\overrightarrow{a_{G3}} \approx \langle -0.515, -0.189 \rangle\ \mathrm{m/s^2}$$

$$\overrightarrow{a_{G4}} \approx \langle -0.257, -0.094 \rangle\ \mathrm{m/s^2}$$

Link 2

$$\sum F_x: \quad A_x + B_x = m_2 a_{G2x}$$

$$\sum F_y: \quad A_y + B_y - W_2 = m_2 a_{G2y}$$

$$\sum M_{G2}: \quad r_{A/G2} \times A + r_{B/G2} \times B + T_2 = 0$$

Link 3 (Extended Coupler)

$$\sum F_x: \quad -B_x + C_x - 4.935 = m_3 a_{G3x}$$

$$\sum F_y: \quad -B_y + C_y - W_3 = m_3 a_{G3y}$$

$$\sum M_{G3}: \quad -0.1524 \sin\theta_3\ B_x + 0.1524 \cos\theta_3\ B_y + 0.752 \sin\theta_3 = I_{G3}\alpha_3$$

Results:

$$\sum F_x: \quad -C_x + D_x = m_4 a_{G4x}$$

$$\sum F_y: \quad -C_y + D_y - W_4 = m_4 a_{G4y}$$

$$\sum M_{G4}: \quad r_{C/G4} \times (-C) + r_{D/G4} \times D = I_{G4}\alpha_4$$

$$A_x = 2.419\ \text{N}, \quad A_y = 8.857\ \text{N}$$

$$B_x = -2.422\ \text{N}, \quad B_y = -8.651\ \text{N}$$

$$B_x = -2.422\ \text{N}, \quad B_y = -8.651\ \text{N}$$

$$C_x = 2.425\ \text{N}, \quad C_y = -7.014\ \text{N}$$

$$D_x = 2.403\ \text{N}, \quad D_y = -6.188\ \text{N}$$

$$T_2 \approx 0.1706\ \text{N·m}$$

These analytical calculations were also supported computationally using MATLAB code. From that code, the following Figures were generated to depict the position, velocity, and acceleration of all the linkages throughout the driven motion of link 2. It is important to note that all values are taken relative to link 1 which in this case is the body of the mechanism. First, Graphs 2.3a through 2.3c depict the position analysis of the mechanism with each number corresponding to its link.

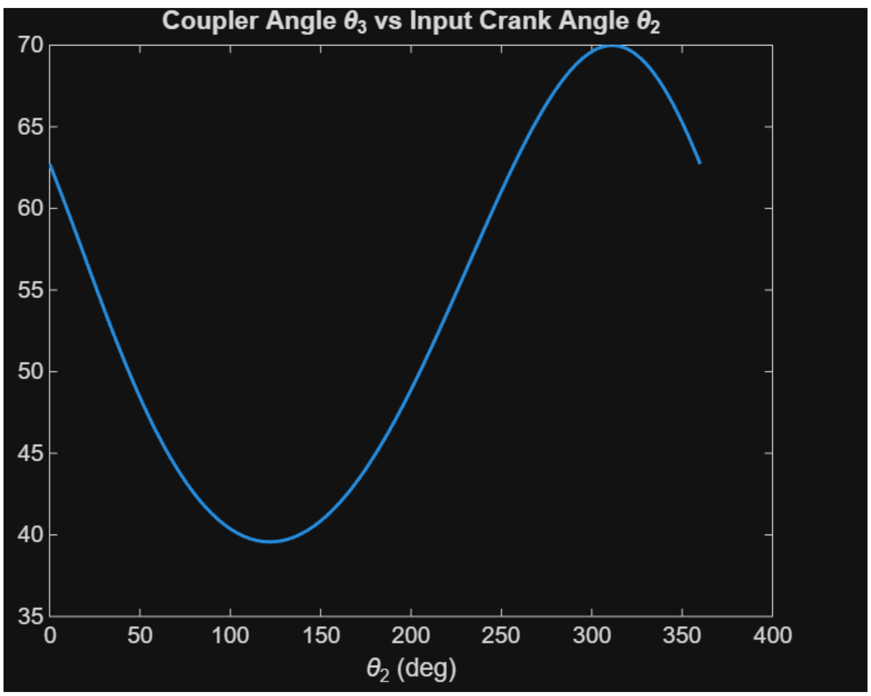

Figure 2.3a: Position Analysis Theta 3 vs Theta 2

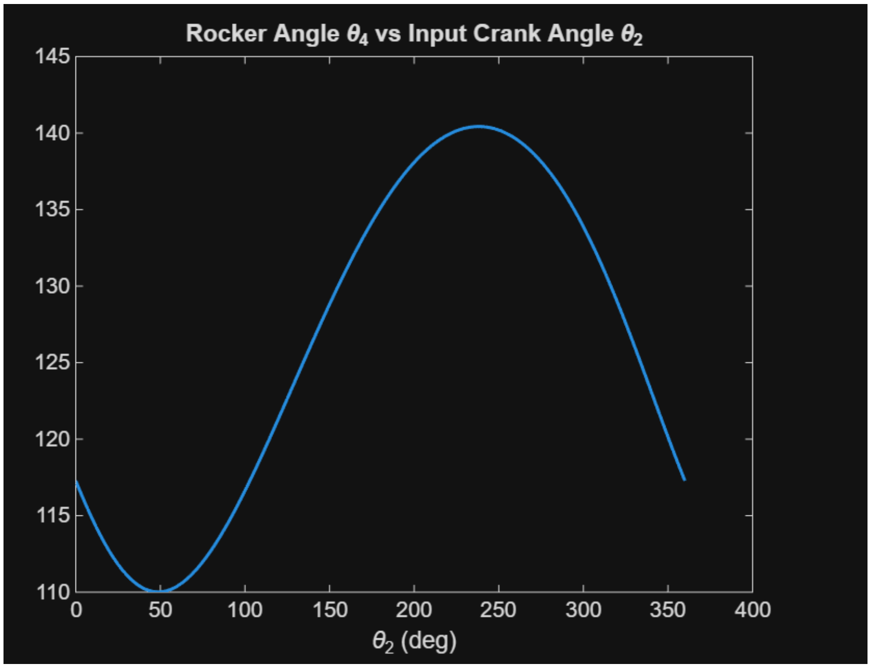

Figure 2.3b: Position Analysis Theta 4 vs Theta 2

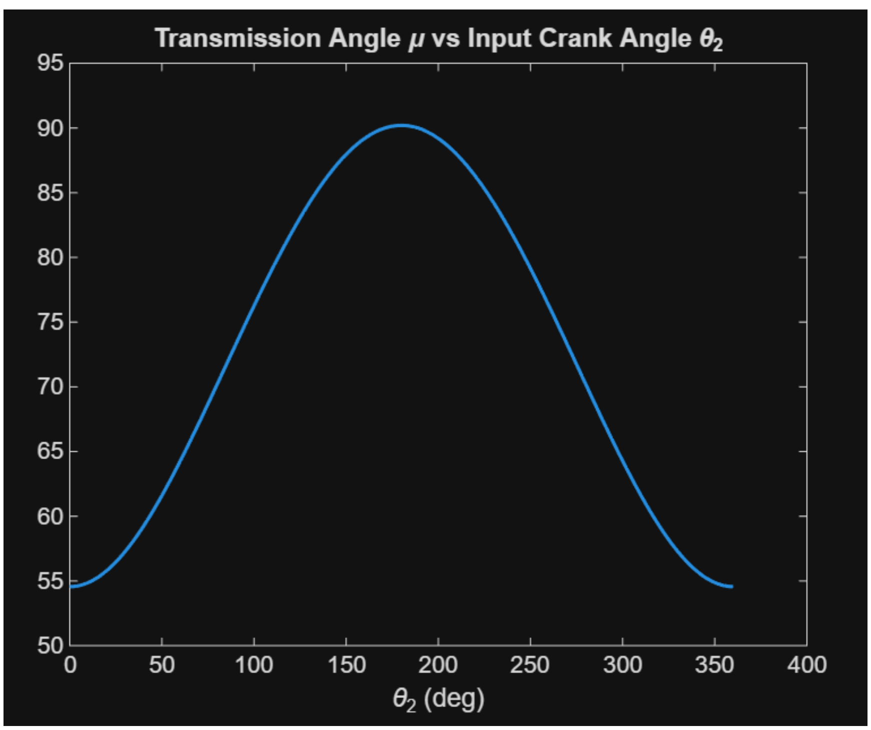

Figure 2.3c: Position Analysis Transmission Angle vs Theta 2

After the position analysis was performed, it was noted that all graphs had smooth transitions throughout the entire rotation of the motor. This meant that the selected link lengths were acceptable in providing motion throughout the entirety of the movement. Next, the velocity analysis was studied to determine if there were any abnormalities which might cause the mechanism to behave irradicably. Figure 2.4 highlights these velocity relationships and Mechanical Advantage with respect to the driving motor.

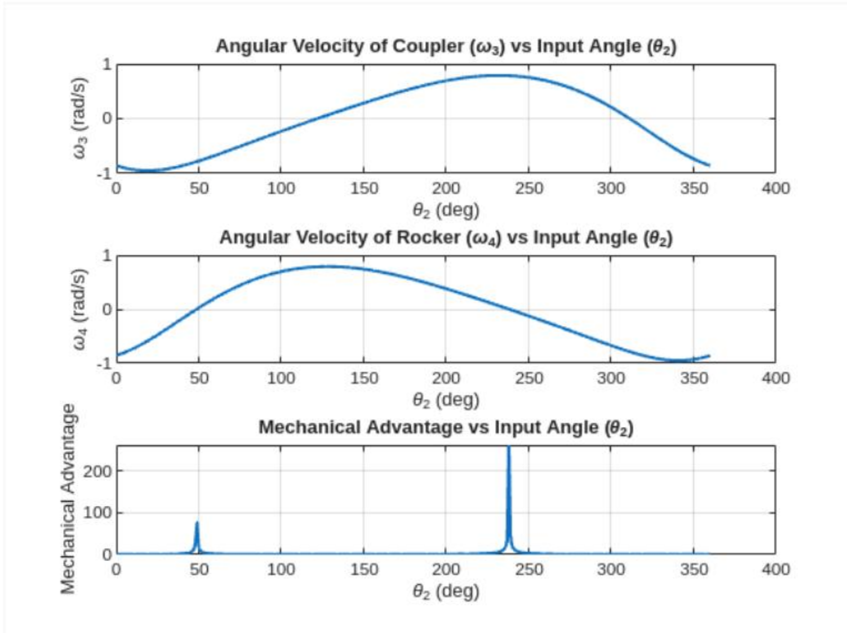

Figure 2.4: Velocity Analysis Omega 2, Omega 4, and Mechanical Advantage vs Theta 2

Lastly, analysis of the acceleration of the different links was carried out. This was to ensure that the motion of the bars was smooth with no sudden jolts. Ultimately, the robot did experience jolts due to the added weight of the microcontroller and payload. However, this would ensure that the rough motion was not due to the geometry of the linkage system. Those accelerations are noted below in Figure 2.5a-c

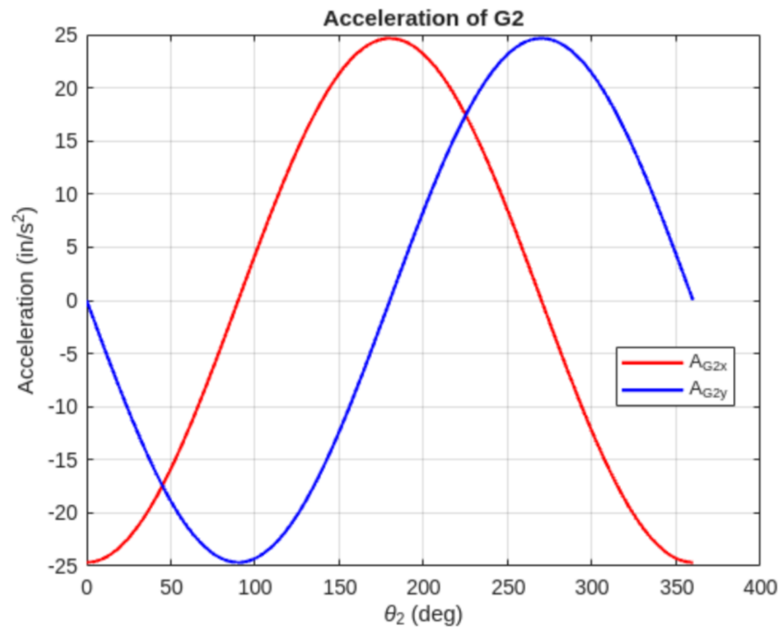

Figure 2.5a: Acceleration of G2

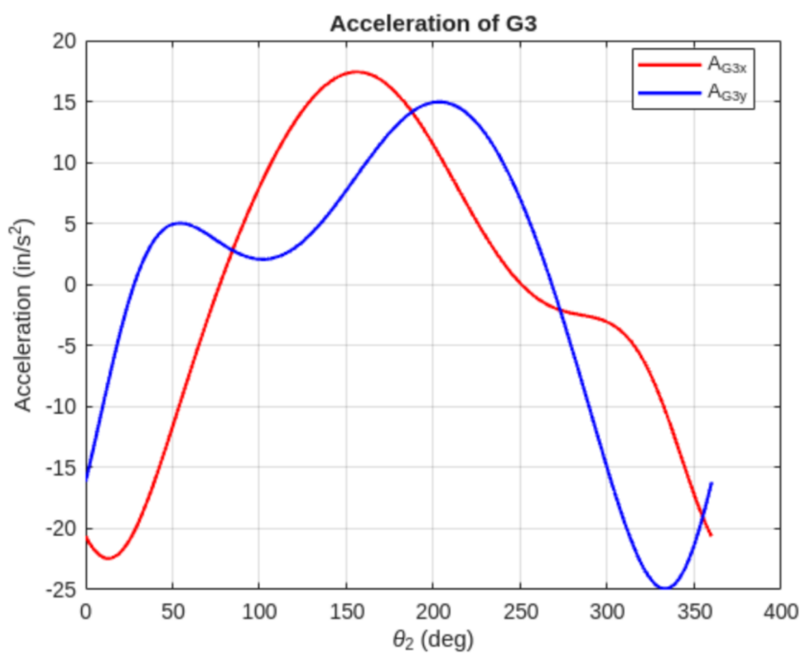

Figure 2.5b: Acceleration of G3

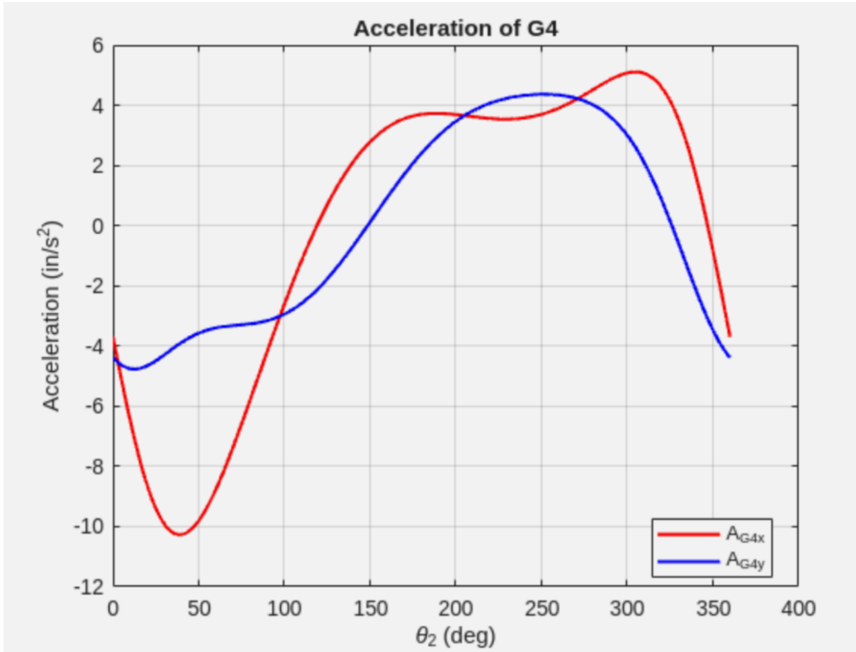

Figure 2.5c: Acceleration of G4

As seen in the figures above, the acceleration is far more unpredictable compared to the positional and velocity analysis portions. This was significant as by looking at the graph, since it was differentiable and continuous, smooth motion was predicted. From this graph, the maximum acceleration of the linkages was also important to study as this value when multiplied with the total mass of the system would account for the force of lift. When compared to the weight of the mechanism, this maximum acceleration should be approximately 32 in/s/s in order to provide a balancing lift force.

2.2.2 Final Design Kinematic Analysis

The final design kinematic analysis was performed to evaluate the performance of the updated linkage geometry and compare it to the initial design. The revised link lengths were selected to improve transmission characteristics while maintaining smooth motion throughout the cycle. All values are reported in SI units.

The updated link lengths used in this analysis are:

$$a = 0.03175\text{m}, \qquad b = 0.13970\text{m}, \qquad c = 0.13970\text{m}, \qquad d = 0.17780\text{m}$$

Using these dimensions, the kinematic analysis was carried out across a full rotation of the input link. The resulting plots, shown in Figures 2.6a through 2.6c, illustrate the position, velocity, and acceleration behavior of the mechanism as functions of θ_2. The curves remain smooth and continuous, indicating that the mechanism satisfies the Grashof condition and operates without locking or singular configurations. Compared to the initial design, the overall motion remains consistent, with only slight shifts in the angular relationships due to the shortened coupler and input link.

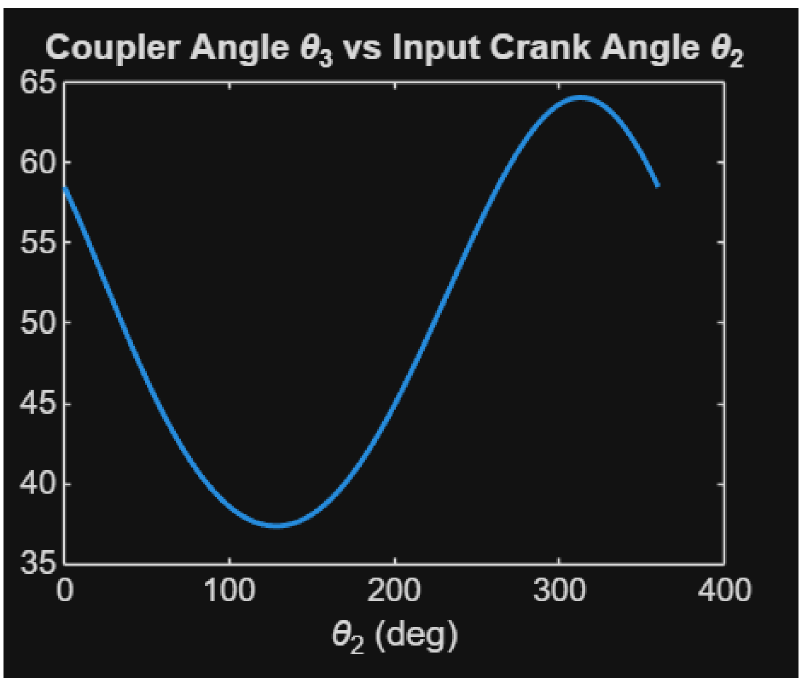

Figure 2.6a: Position Analysis Theta 3 vs Theta 2

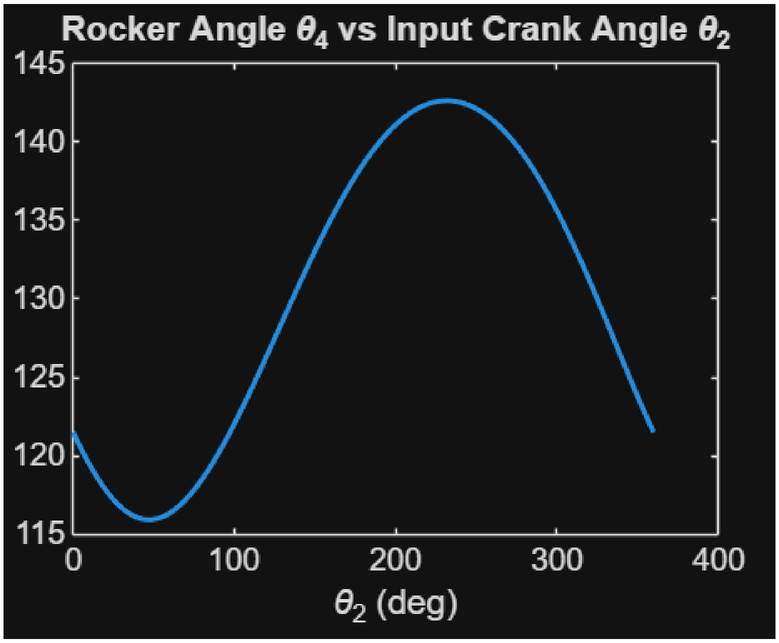

Figure 2.6b: Position Analysis Theta 4 vs Theta 2

The transmission angle was also evaluated for the updated geometry. The results show an overall improvement in the minimum transmission angle compared to the initial configuration, indicating more effective force transmission throughout the motion cycle. This improvement suggests that the final design is better suited for load-bearing applications such as climbing, as it reduces the likelihood of inefficient force transfer at critical positions.

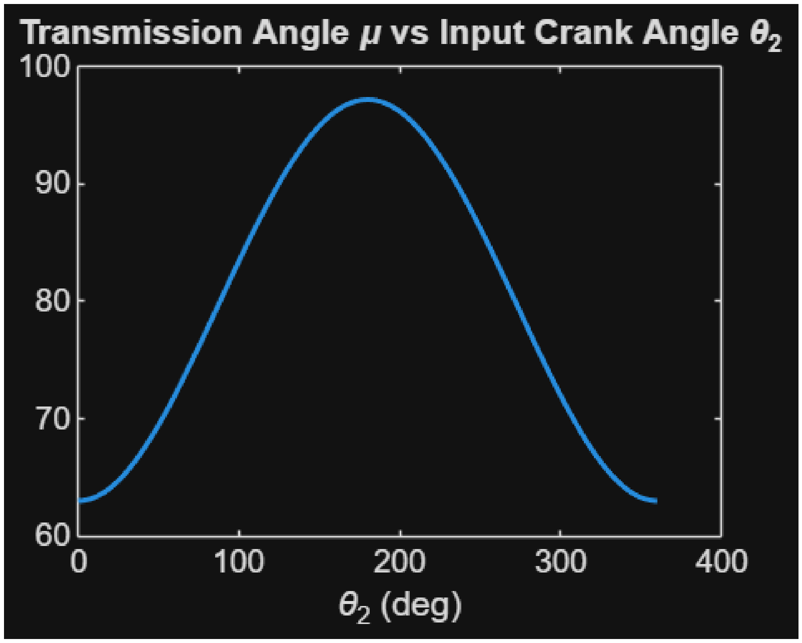

Figure 2.6c: Position Analysis Transmission angle vs Theta 2

Velocity analysis was performed to determine the angular velocities of links 3 and 4. The results, as seen below in Fig. 2.7, indicate smooth velocity profiles with no abrupt changes, confirming stable motion behavior. While peak angular velocities are slightly altered due to the reduced link lengths, the overall trend remains similar to the initial design. This consistency ensures that the mechanism will not introduce unexpected dynamic effects during operation.

The mechanical advantage was also examined using the angular velocity relationships between the links. The updated design demonstrates a more favorable distribution of mechanical advantage, particularly in regions where lifting or pulling forces are required. This aligns with the intended purpose of the mechanism, as improved mechanical advantage directly contributes to more efficient climbing performance.

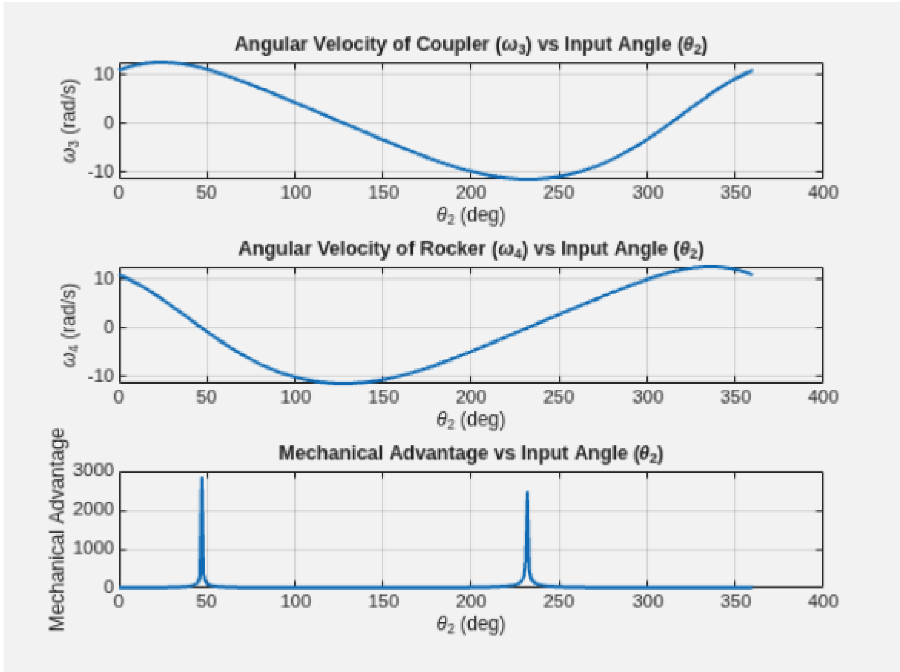

Figure 2.7: Velocity Analysis Omega 2, Omega 4, and Mechanical Advantage vs Theta 2

Finally, acceleration analysis was conducted to assess the dynamic behavior of the system. The results, as shown in Figures 2.8a through 2.8c, exhibit greater variability compared to the position and velocity plots. However, the curves remain continuous and differentiable, indicating that the motion is still smooth and free of kinematic discontinuities. Any observed spikes in acceleration are attributed to geometric effects rather than instability in the mechanism.

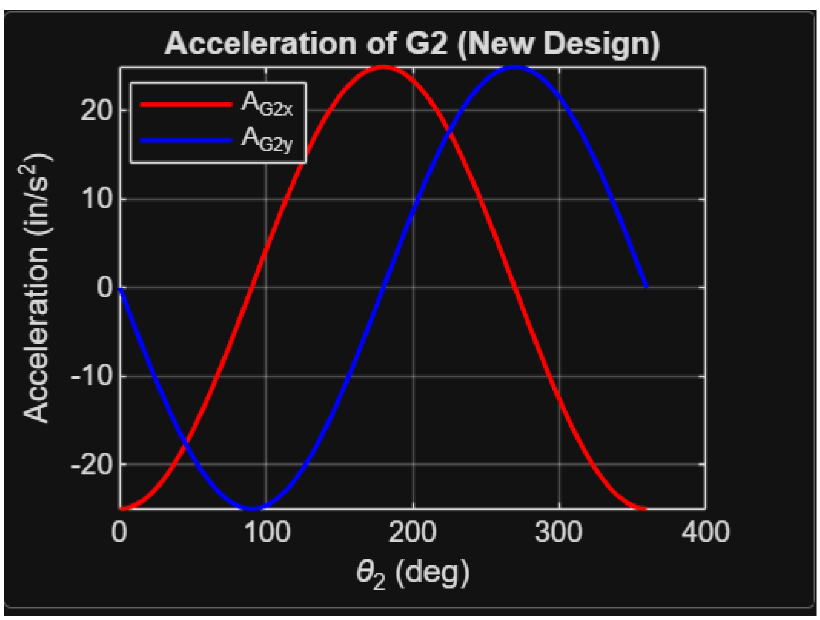

Figure 2.8a: Acceleration of G2

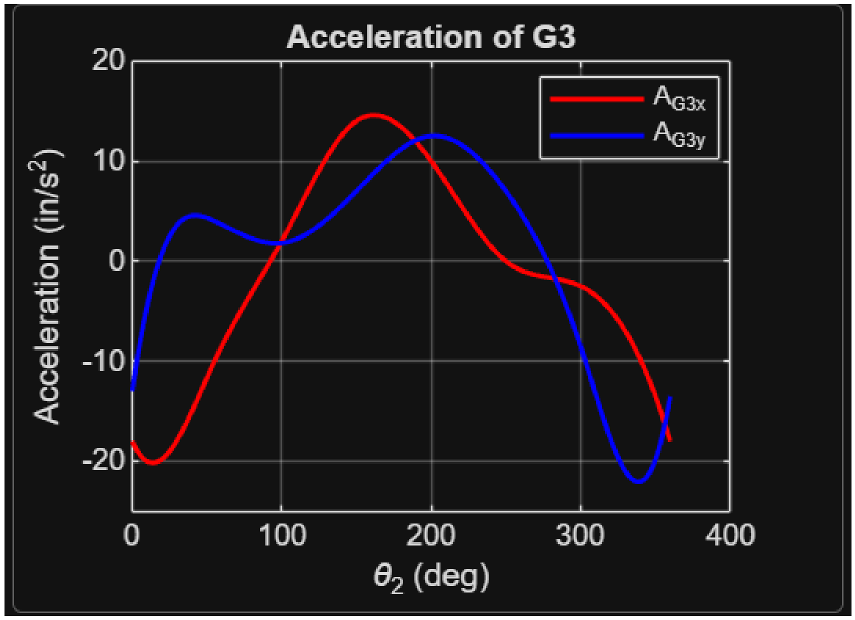

Figure 2.8b: Acceleration of G2

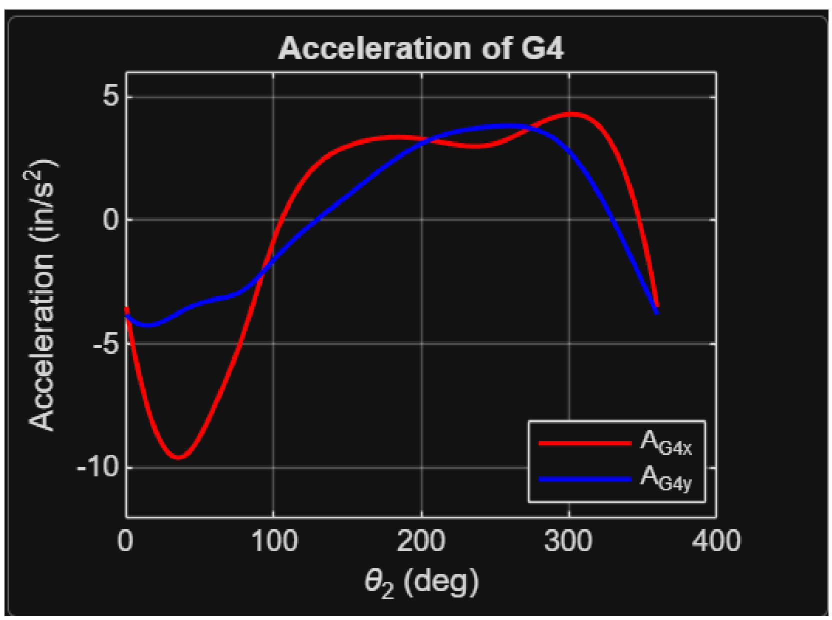

Figure 2.8c: Acceleration of G2

Overall, the final design maintains the desirable smooth motion characteristics of the initial configuration while improving force transmission and mechanical advantage. These enhancements make the updated linkage geometry more effective for the intended climbing application, particularly under load conditions where efficient force transfer is critical.

2.2.3 Joint analysis

Using the joint reaction forces obtained from the dynamic joint analysis, the required input torque was determined by summing moments about the center of mass of link 2. The joint reaction forces at A, B, C, and D were first determined using dynamic equilibrium equations for each link, as shown in Section 2.2.1. These values were then used to compute the required input torque.

The moment equation for link 2 is:

$$r_{\frac{A}{G2}} \times \mathrm{A} + r_{\frac{B}{G2}} \times \mathrm{B} + \mathrm{T} = 0$$

Solving for the input torque gives:

$$T = -\left(r_{\frac{A}{G2}} \times A + r_{\frac{B}{G2}} \times B\right)$$

Substituting the joint reaction forces from the dynamic analysis gives:

$$\mathrm{T} \approx 0.1706\, N \cdot m$$

This value represents the torque required at the input link to drive one side of the four-bar linkage under the applied loading condition. Since the robot uses two four-bar mechanisms, this torque value was important for determining whether the selected motor and gear system could provide enough output torque for the final design.

Although this analysis was performed at $\theta_2 = 45°$, it was later recognized that the joint analysis should have been conducted at the position corresponding to the minimum transmission angle, where the mechanism experiences its least efficient force transfer. At this configuration, the required input torque increases to approximately 0.367 N·m. This discrepancy helps explain the poorer-than-expected performance observed during testing, as the motor and gear system were not adequately sized to handle the higher torque demands present at critical positions in the motion cycle.

Following the joint force analysis, the factor of safety for the primary joints was evaluated using the reaction forces calculated at each connection point. The resultant force at each joint was determined from the x- and y-components, and the corresponding shear stress was computed assuming double shear conditions.

Pin Diameter = ¼ in

Double Shear: $\tau = \frac{F}{2A}$

Area = $\pi \cdot r^2 = 3.17 \times 10^{-5}\, m^2$

$\tau_{allowable} = 145MPa$

Joint A:

$$F_A = \sqrt{2.419^2 + 8.857^2} = 9.18\, N$$

$$\tau_A = \frac{9.18}{2(3.17 \times 10^{-5})} = 0.145MPa$$

$$FOS_A = \frac{145}{0.145} = 1000$$

Joint B:

$$F_B = \sqrt{-2.422^2 + 8.651^2} = 8.98\, N$$

$$\tau_B = \frac{8.98}{2(3.17 \times 10^{-5})} = 0.142MPa$$

$$FOS_B = \frac{145}{0.142} = 1020$$

Joint C:

$$F_C = \sqrt{2.425^2 - 7.014^2} = 7.42\, N$$

$$\tau_C = \frac{7.42}{2(3.17 \times 10^{-5})} = 0.117MPa$$

$$FOS_C = \frac{145}{0.117} = 1240$$

Joint D:

$$F_D = \sqrt{2.403^2 - 6.188^2} = 6.64\, N$$

$$\tau_D = \frac{6.64}{2(3.17 \times 10^{-5})} = 0.105MPa$$

$$FOS_D = \frac{145}{0.105} = 1380$$

The lowest factor of safety occurs at joint A, with FOS ≈ 1000. Since all factors of safety are much greater than 1, the 1/4 in steel pins are highly overdesigned for the expected joint loads. Therefore, failure at the joints was not expected to be a limiting factor in the mechanism.

2.3 Power Transmission

One of the key aspects to providing enough torque to the 4-Bar linkage system is using power transmission. By itself, the motor can provide a predetermined amount of torque at a specified angular velocity. Although the total power must be conserved from the motor to the linkage system, the amount of torque provided may be scaled at the cost of the angular velocity of the output. To determine the amount of torque the robot would require moving the linkage system, a dynamics simulation was utilized, and the corresponding maximum torque applied was calculated.

2.3.1 Altair Inspire Dynamics Simulation

An Altair Inspire Motion model was developed to simulate the kinematic and dynamic behavior of the final mechanism design. The purpose of this model was to verify that the mechanism operates as intended and to evaluate system motion, loads, and motor requirements.

The model was constructed with the following key elements:

> **Joint Definitions:** All connections between links were defined using appropriate joint types (primarily revolute joints) to accurately represent the physical pivot behavior of the mechanism.
>
> **Ground Definition:** The base frame of the mechanism was fixed to serve as the ground link, ensuring proper constraint of the system.
>
> **Motor Input:** Two rotational motors were applied within the simulation to drive the linkage system. This approach was used to simplify the model and avoid explicitly modeling the gear reduction system in the software.
>
> **Link Properties:** All components were assigned approximate material properties and masses to enable realistic dynamic analysis.

Although two motors were used in the simulation, the final physical design will utilize a single motor with a gear reduction system to distribute motion appropriately. The dual-motor setup in the model serves as a simplification to replicate the intended motion without increasing model complexity.

A motion simulation was then performed to observe the operation of the mechanism over a full cycle. The simulation confirmed that the linkage moves smoothly without unintended constraints, interference, or unrealistic motion.

Additionally, the software was used to estimate motor torque requirements and system loads, which were later used to validate the feasibility of the selected motor and gear reduction system.

A short animation of the simulation was submitted separately to visually demonstrate the mechanism's operation and verify proper functionality.

2.3.2 Calculations

Based on the results of the dynamic force analysis, the required input torque exceeds the rated torque output of the specified motor. As a result, the motor alone is insufficient to drive the mechanism under the given loading conditions. To address this limitation, a gear reduction system will be incorporated to increase the available output torque. By selecting an appropriate gear ratio, the motor torque can be amplified to meet the required demand, while accounting for potential efficiency losses within the transmission. This modification will ensure that the mechanism operates reliably under the applied load.

Quantitatively, a gear ratio of 1:1.25 should be enough to power the entire mechanism. The main issue behind using this value is, based on the total stroke length of the 4-bar mechanism, the robot will have a difficult time traversing the rope in under a minute, which was one of the requirements of the project.

2.4 Drive Shaft

The drive shaft is a critical component responsible for transmitting torque from the motor and bevel gear system to the four-bar linkage. In addition to torsional loading from the motor, the shaft experiences bending due to forces generated at the gear mesh. Because the shaft supports both the mounted gear and the linkage connections, it is subjected to combined loading conditions that must be evaluated to ensure structural integrity.

To verify the reliability of the design, the shaft was analyzed using both analytical failure theory and finite element analysis (FEA). The analytical approach evaluates stresses due to torsion and bending and applies the von Mises failure criterion to determine the factor of safety. This is followed by a numerical simulation to validate the analytical results and confirm that the shaft can safely withstand the expected operating loads.

2.4.1 Failure Theory and Safety Factor

The drive shaft was analyzed to determine whether it could safely transmit the required torque from the bevel gear to the linkage system. The shaft was made from aluminum, while the mounted bevel gear was made from PLA. The 16-tooth bevel gear was mounted on the shoulder of the larger center section of the shaft. Since the smaller shaft diameter is the weakest section, the shaft was analyzed using the smaller diameter of d = 5/16 in.

The required shaft torque from the dynamic analysis was T = 0.1706 N·m

Converting to English units: T = 1.51 lbf·in

$$d_p = \frac{N}{P_d}$$

$$d_p = \frac{16}{8}$$

$$d_p = 2.00 \text{ in}$$

$$r_p = 1.00 \text{ in}$$

The tangential force applied to the gear was calculated as:

$$F_t = \frac{T}{r_p}$$

$$F_t = \frac{1.51}{1.00}$$

$$F_t = 1.51 \text{ lbf}$$

Since the gear is mounted against the shoulder of the shaft, the bending moment was approximated using half of the gear face width as the moment arm:

$$L = \frac{0.50}{2} = 0.25 \text{ in}$$

$$M = F_t L$$

$$M = 1.51(0.25)$$

$$M = 0.378 \text{ lbf} \cdot \text{in}$$

The torsional shear stress in the shaft was calculated using:

$$\tau = \frac{16T}{\pi d^3}$$

$$\tau = \frac{16(1.51)}{\pi(0.3125)^3}$$

$$\tau = 252 \text{ psi}$$

The bending stress in the shaft was calculated using:

$$\sigma = \frac{32M}{\pi d^3}$$

$$\sigma = \frac{32(0.378)}{\pi(0.3125)^3}$$

$$\sigma = 126 \text{ psi}$$

The combined stress was found using the von Mises failure theory:

$$\sigma_{vm} = \sqrt{\sigma^2 + 3\tau^2}$$

$$\sigma_{vm} = \sqrt{(126)^2 + 3(252)^2}$$

$$\sigma_{vm} = 456 \text{ psi}$$

Using conservative aluminum yield strength of:

$$S_y = 10000 \text{ psi}$$

The factor of safety was:

$$n = \frac{S_y}{\sigma_{vm}}$$

$$n = \frac{10000}{456}$$

$$n = 21.9$$

Based on this result, the aluminum drive shaft has a high factor of safety under the expected loading conditions. The calculated von Mises stress is much lower than the conservative aluminum yield strength, indicating that yielding of the shaft is unlikely during operation. Therefore, shaft failure was not expected to be a limiting factor in the final design.

2.4.3 Finite Element Analysis

A finite element analysis (FEA) was performed in Abaqus/CAE to validate the analytical stress calculations for the drive shaft. The shaft was modeled as a three-dimensional deformable solid with aluminum material properties and meshed using tetrahedral elements (C3D10).

Both ends of the shaft were fixed to represent support conditions. A torque of 0.1706 N·m was applied about the shaft axis using a reference point and kinematic coupling, and a tangential gear force of 6.72 N was applied at the gear mounting location.

The resulting von Mises stress distribution is shown in Figure X. The maximum stress occurred near the smaller diameter region and geometric transitions, consistent with expected stress concentration behavior. The maximum von Mises stress from the simulation was approximately 830 Pa (0.120 psi), which is significantly lower than the analytical result of 456 psi. This difference is due to unit scaling inconsistencies from importing inch-based geometry into Abaqus while using SI units for loads and material properties.

The deformation shown in the FEA results is highly exaggerated due to a large deformation scale factor applied by Abaqus for visualization (on the order of 10^7). The actual deformation of the shaft is negligible. Overall, the FEA confirms that the shaft experiences very low stress and is not expected to fail under the given loading conditions.

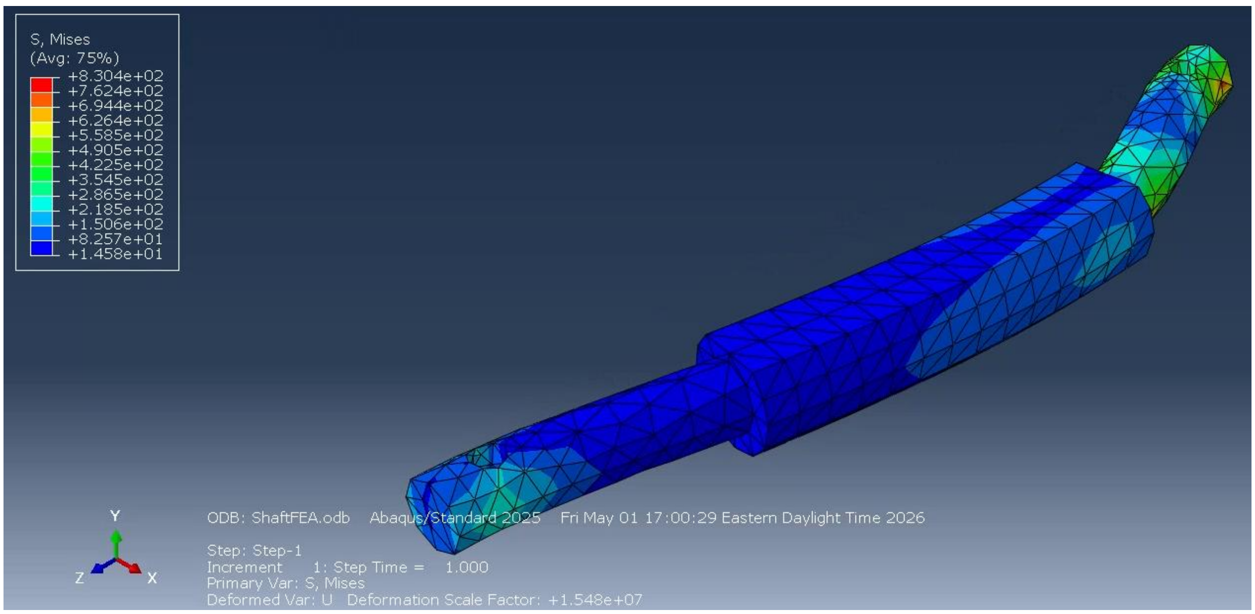

Figure 2.9: Von Mises stress distribution in the drive shaft from Abaqus FEA

3.0 MANUFACTURING

The fabrication of the robot components relied on a combination of conventional machining processes to achieve both accuracy and structural reliability. Basic structural elements, such as the linkages, were manufactured using a band saw for rough cutting and a drill press for producing precise holes required for pin joints and fasteners. These methods allowed for efficient material removal while maintaining sufficient dimensional control to ensure proper alignment and smooth motion within the mechanism.

More precise components were produced using a lathe and milling machine, particularly for the axle that drives link 2. The lathe was used to turn the axle to the required diameter and ensure concentricity, which is critical for minimizing vibration and ensuring consistent rotational motion. Milling operations were then applied to refine features such as flats or keyways, enabling secure coupling between the motor and the linkage system. The integration of these manufacturing techniques allowed for a balance between rapid fabrication and precision, resulting in a functional and reliable mechanical system.

3.1 Bill of Materials

Since the project team had minimal prior manufacturing experience, there were several materials that needed to be ordered using the Mechanical Engineering Department's budget. The team was given a budget of one hundred dollars to purchase any equipment necessary for the project. Below, Table 3.1 breaks down the expenses the team encountered.

Table 3.1: Purchased Materials.

Item	Cost
3D Printer Filament	\$52.49
Metal Pins	\$5.89
Non-Slip Surface Grip Pads	\$13.47
E6000 Craft Adhesive	\$8.09
Total	79.94

The above items were purchased for the project ultimately enhancing the overall design. 3D Printer Filament was purchased by the group as Kopiwoda had access to a personal printer, which was used throughout the design process. This filament was stronger than the PVC filament that we already had. Initial prototype designs experienced flexing, demonstrated acting stresses which could ultimately lead to failure. Making the body, and arms of the design out of this filament increased the amount of stress the mechanism was able to take, which allowed it to function at the highest capacity.

Metal pins were ordered from Amazon.com to serve as joints between the different linkages. In the end, these pins were not successfully delivered, as Amazon.com mixed up the orders and sending a bike mounted light instead. As a work around, the team purchased lock nuts out of pocket at a local hardware store and used ¼ 20 bolts as the pins. These pins were over engineered, having a factor of safety of larger than 10, meaning that failure at the joints was highly unlikely.

Non-Slip Surface Grip Pads were purchased to be attached to robotic arms. In previous tests, there was a noticeable amount of slipping between the plastic arm and the nylon rope. A large portion of the stroke length of the mechanism was lost due to this slipping. To combat the slipping, these grip pads were added to the arms. This increased the friction between the arm and the rope leading to better performance.

Lastly, E600 Craft Adhesive was purchased to glue the arms of the robot onto the linkages. This ended up being unnecessary as the geometry of the arms was constantly being iterated on. Therefore, it did not make a lot of sense to use glue, except on the final design. It is important to note that there was another plastic pin mechanism that was created to easily swap the design of the arm being tested. Therefore, before the final design presentation during Design Day, it became unnecessary to use the craft adhesive glue.

3.2 Prototyping Modifications and Overall Design

Throughout the prototyping phase, several modifications were made to improve the performance, manufacturability, and functionality of the mechanism. While the initial CAD model provided a strong starting point, it did not include dimensions for link 1 which would include space for the motor, nor did it fit within the "shoebox" requirements. Once a prototype was constructed based on that model, limitations in alignment, speed, and grip were discovered and required corrections.

3.2.2 Body Sizing

One of the primary changes made during prototyping was reducing the overall size of the mechanism. By rotating the motor 90 degrees, the width of Link 1 (the main body) could be decreased to create a more compact design that better met the project size constraints and improved balance during traversal. A smaller body also reduced unnecessary weight, which helped decrease the load on the motor and linkage system. This modification contributed to smoother motion and reduced the likelihood of tipping during operation.

3.2.3 Link Lengths

The lengths of all linkage members were shortened from the initial design. This adjustment was made to improve control of the mechanism and reduce excessive motion that was observed during early testing. Shorter links resulted in a more compact stroke and allowed for better force transmission through the system. It also allowed the design to fit snuggly into the "shoebox" requirements.

Additionally, Link 2 was thickened so that the axel could screw directly into the piece, preventing any rotational differences between the two pieces. While this increased the width of the project, the thinning of Link 1 more than made up for the gain in width caused by Link 2.

3.2.4 Gear Orientation

A major modification to the power transmission system was the change in both gear type and gear ratio. The initial prototype utilized a herringbone gear configuration with a 12-tooth pinion and 30-tooth gear, resulting in an approximate gear ratio of 1:2.5. This setup provided a higher torque output at the expense of rotational speed.

During the redesign process, the motor was rotated 90 degrees to better integrate within the compact body. This change required a different gear configuration, leading to the implementation of bevel gears. The final design used a 12-tooth pinion and a 16-tooth gear, resulting in a reduced gear ratio of approximately 1:1.25. This configuration allowed for a more direct transmission of motion and improved packaging within the system.

While the bevel gear setup increased the output speed of the mechanism, it significantly reduced the available torque compared to the original design. As a result, although the final system demonstrated faster motion during testing, it was unable to consistently generate sufficient torque to overcome the resistance forces from the rope and linkage system. This limitation ultimately led to failure during operation, as the mechanism could not maintain reliable traversal under load.

This trade-off highlights the importance of balancing speed and torque in power transmission design. Future iterations should consider increasing the gear reduction ratio or selecting a motor with a higher torque output to ensure adequate performance.

3.2.5 Axle Length

The axle length was reduced from the initial design to match the narrower body and updated linkage spacing. A shorter axle minimized unnecessary overhang, which prevented Link 3 from colliding with the axel during its rotation. This change, combined with spacers that were created for link 4, helped maintain proper alignment between the linkage arms on both sides of the mechanism.

4.0 EXPERIMENTAL RESULTS

4.1 Design Day

During Design Day testing, the performance of the rope-traversing mechanism was limited by both structural and power transmission issues. Prior to testing, the final iteration of the mechanism experienced a failure in the gripping components ("hands"), which broke during assembly. As a result, the team was required to revert to a previous design of the gripping system. This earlier configuration provided reduced contact and less effective engagement with the rope, leading to insufficient traction during operation.

Due to the decreased grip, the mechanism experienced noticeable backward slipping during traversal attempts. Instead of maintaining consistent forward progress, the device would intermittently lose traction and slide along the rope, reducing overall efficiency and preventing successful completion of the task.

In addition to the gripping issue, the system also suffered from insufficient torque output. The updated gear configuration, which prioritized speed through a lower gear reduction ratio, limited the torque delivered to the linkage system. As a result, the motor was unable to consistently drive the mechanism, both under load and in no-load conditions. During testing, the system frequently stalled and became stuck at certain positions in the motion cycle, indicating that the available torque was not sufficient to overcome the peak dynamic and frictional forces within the mechanism.

These combined issues ultimately prevented the mechanism from achieving reliable rope traversal during Design Day.

Despite these shortcomings, the results provided valuable insight into the design trade-offs within the system. The faster motion achieved with the updated gear ratio demonstrated the potential for improved traversal speed; however, it came at the cost of torque, which proved to be the limiting factor.

5.0 CONCLUSION

5.1 Marketing Considerations

The prototype mechanism was fabricated using white filament, while the intended final version was produced in black filament. However, due to component failures encountered on Design Day, the final assembly was constructed using a combination of both black and white printed parts. As a result, the team shifted the design theme from a monkey/jungle concept to a cow-themed mechanism, which was completed with a small foam cow mascot positioned near the motor.

5.2 Future Improvements

Future improvements would focus on optimizing the linkage geometry to produce larger step sizes per cycle while maintaining smooth motion, as well as increasing the gear reduction ratio or selecting a higher-torque motor. With these adjustments, the mechanism would be better equipped to generate sufficient driving force and maintain consistent grip, ultimately improving overall performance.

APPEDIX – CAD DRAWINGS

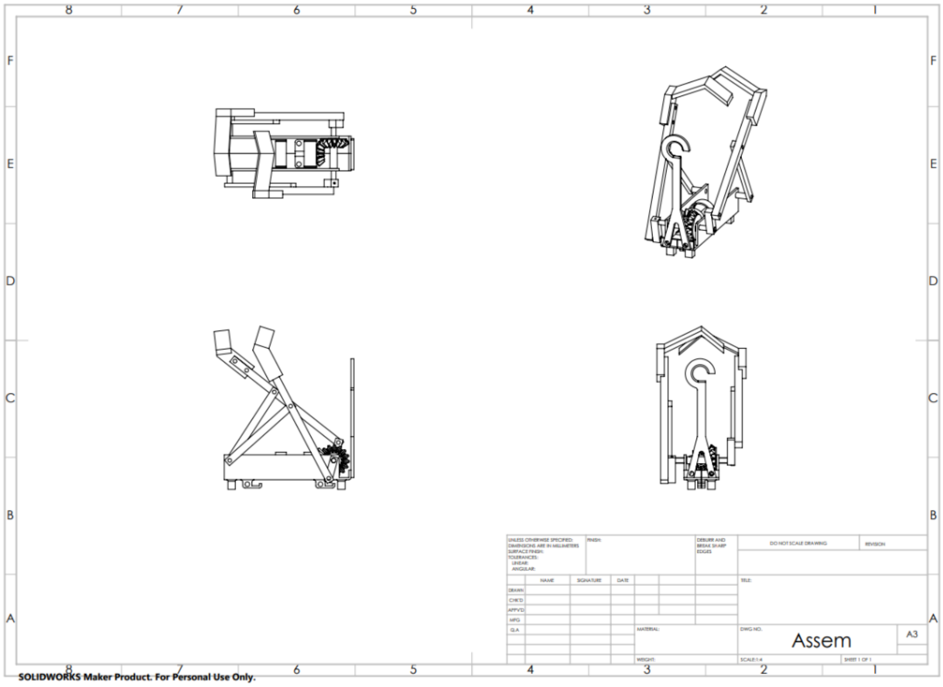

Figure A.1: Full assembly drawing

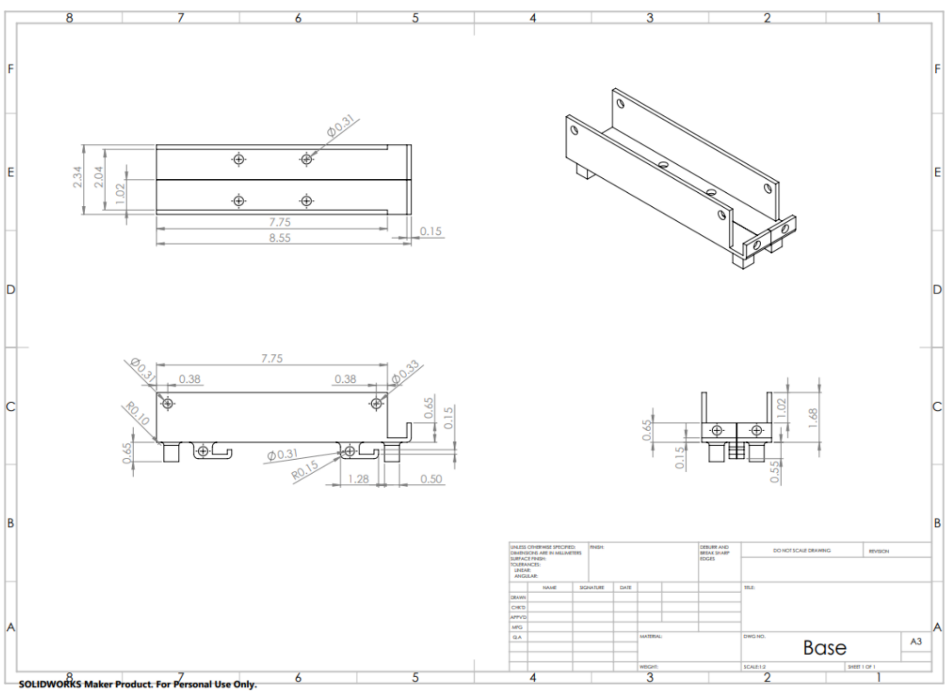

Figure A.2: Base (link 1) drawing with dimensions

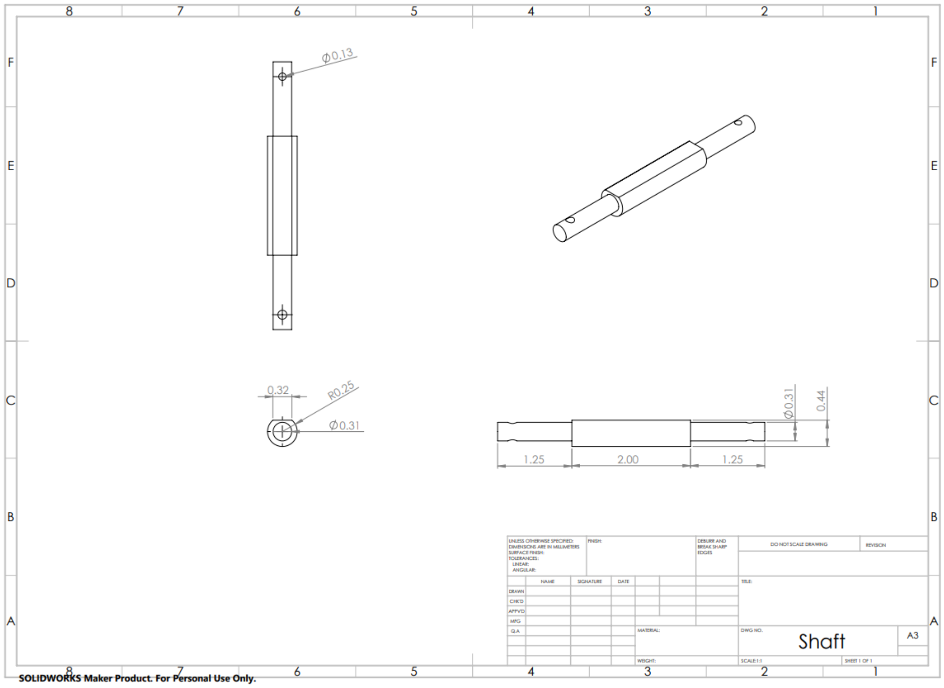

Figure A.3: Shaft drawing with dimensions

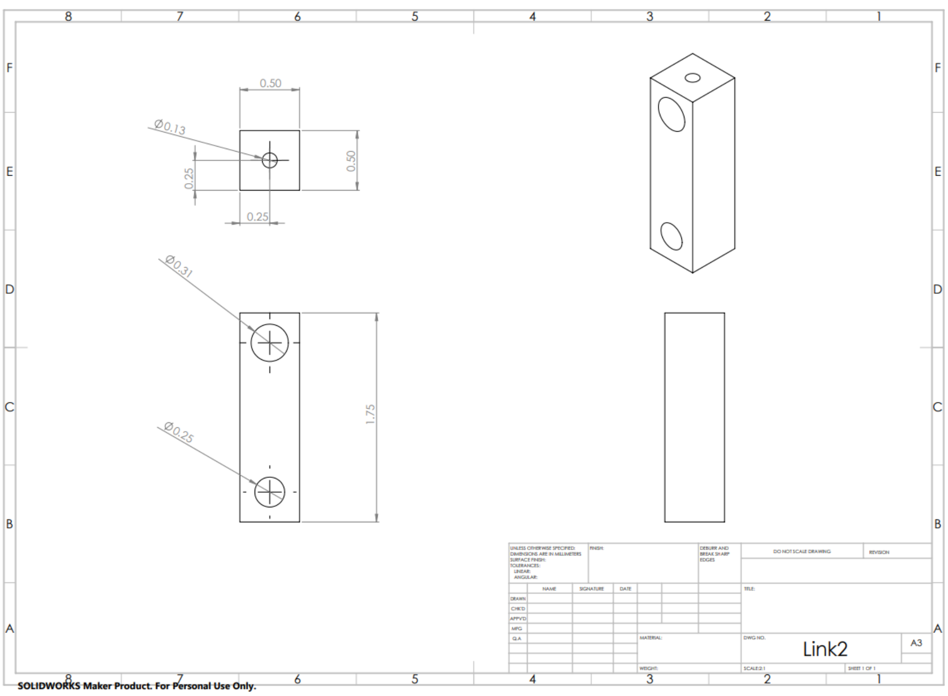

Figure A.4: Link 2 drawing with dimensions

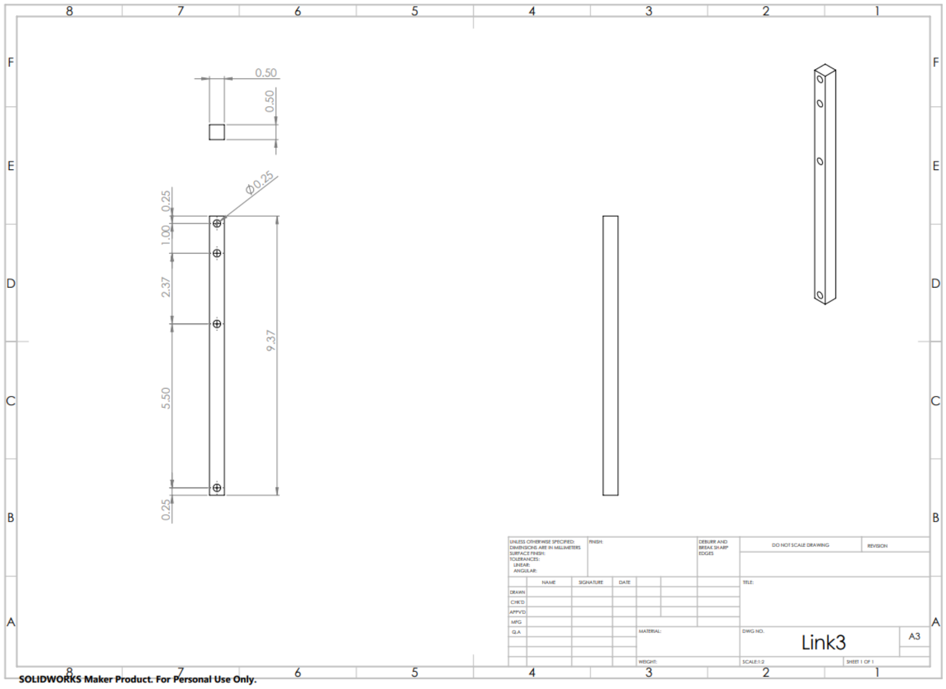

Figure A.5: Link 3 drawing with dimensions

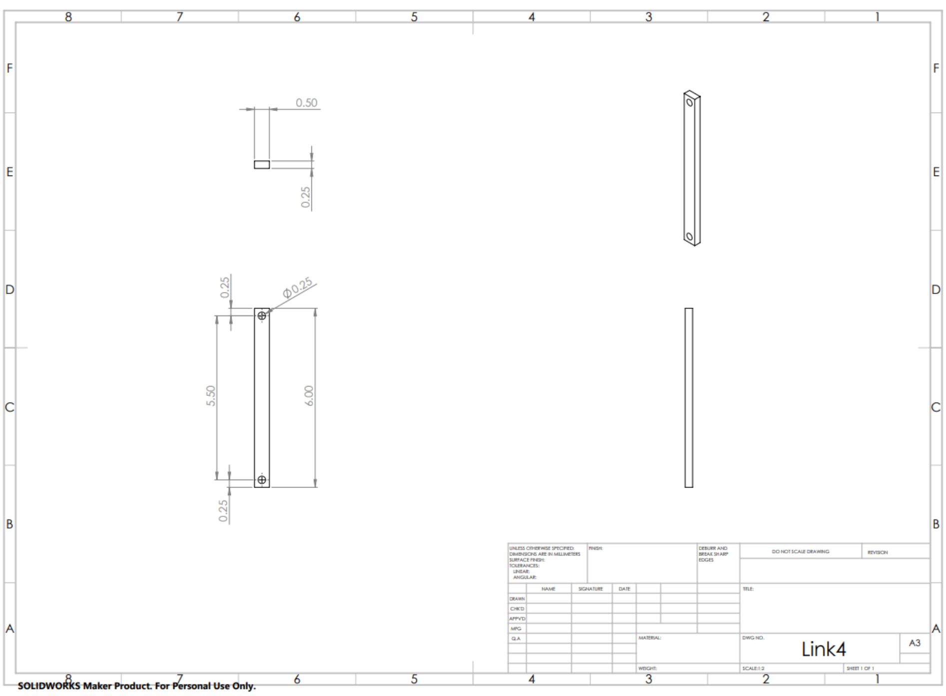

Figure A.6: Link 4 drawing with dimensions

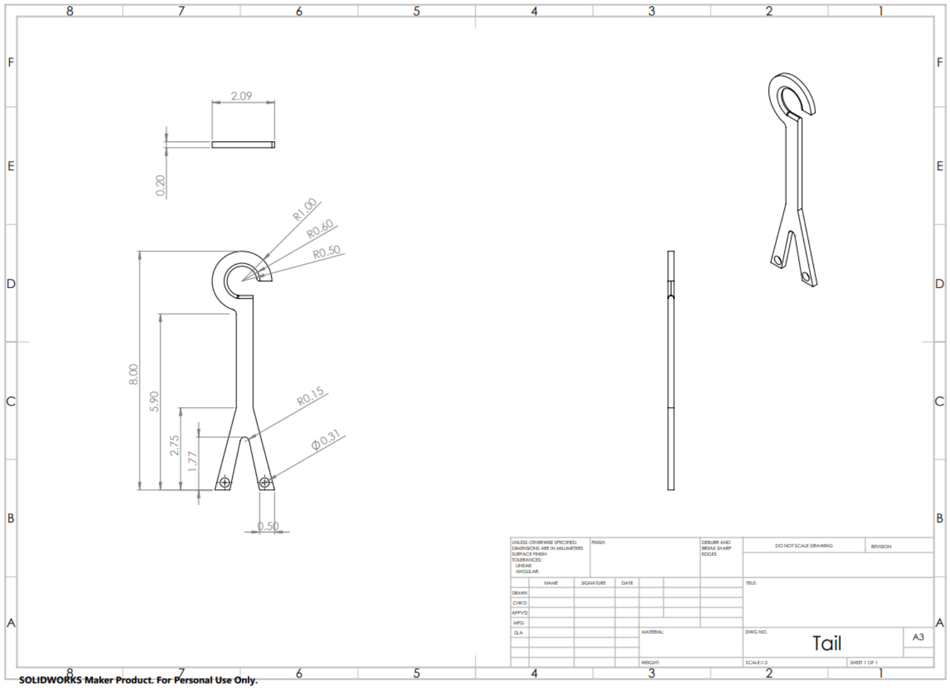

Figure A.7: Tail drawing with dimensions

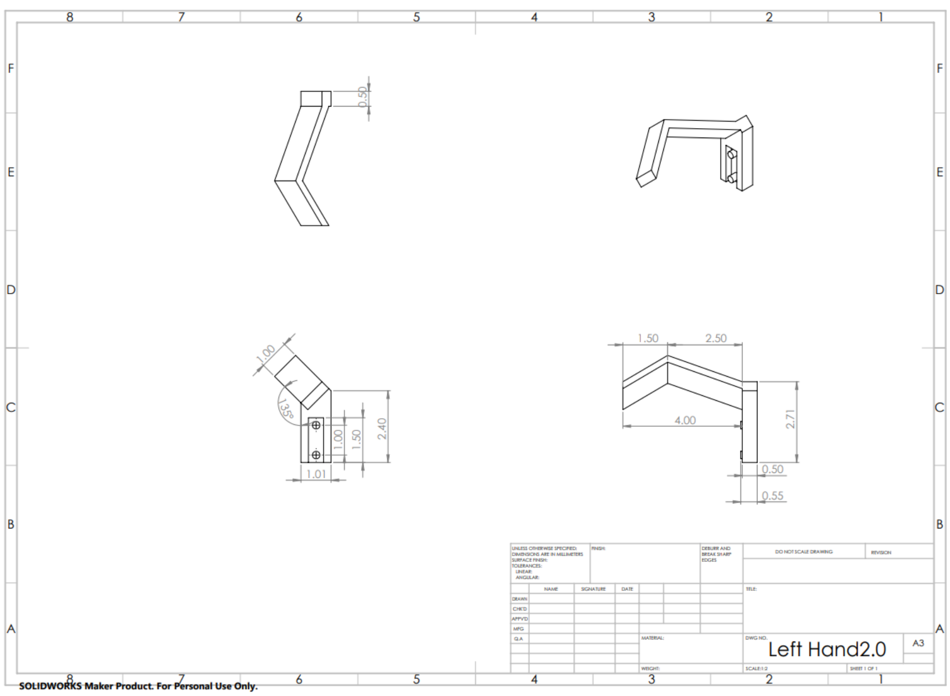

Figure A.8: Right hand drawing with dimensions

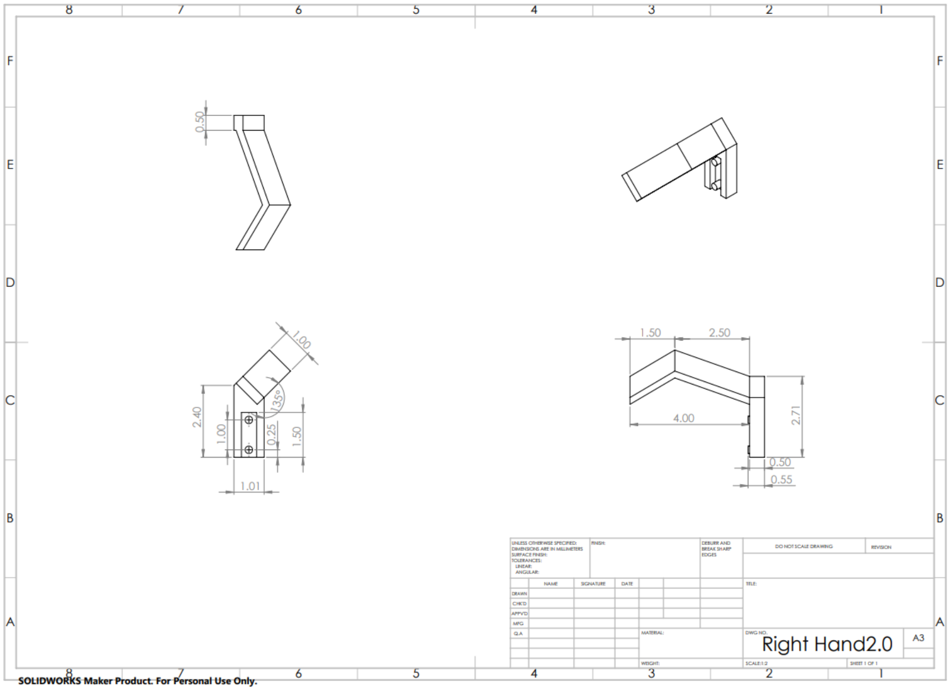

Figure A.9: Left hand drawing with dimensions

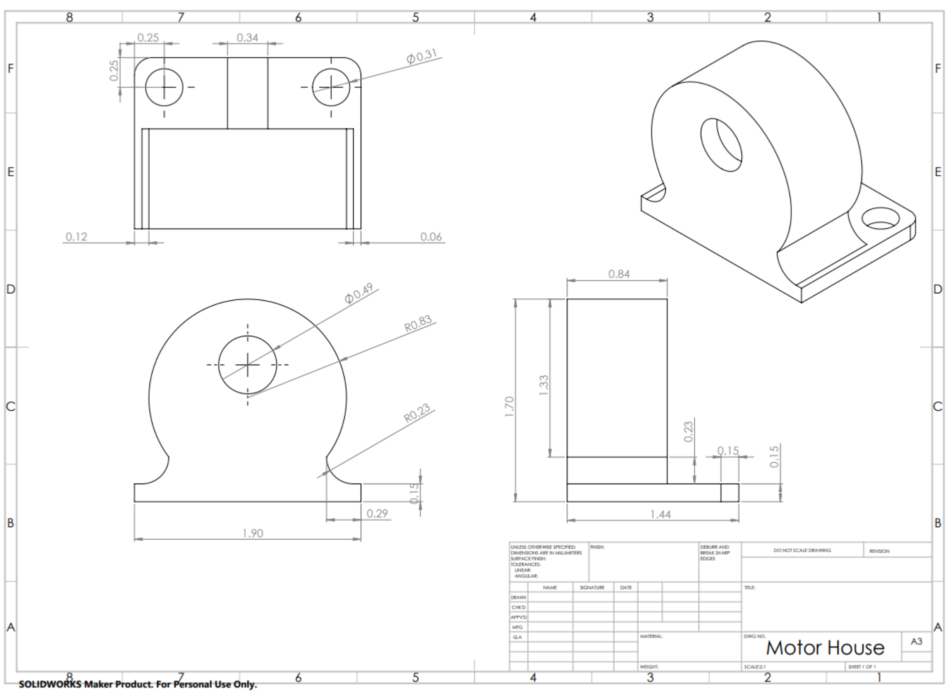

Figure A.10: Motor house front drawing with dimensions

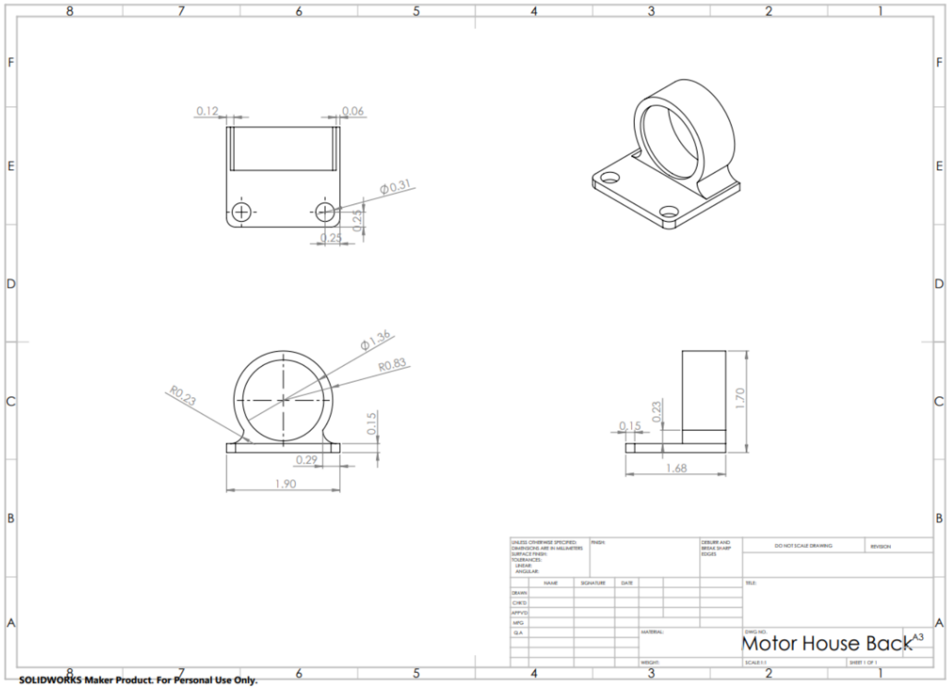

Figure A.11: Motor house back drawing with dimensions

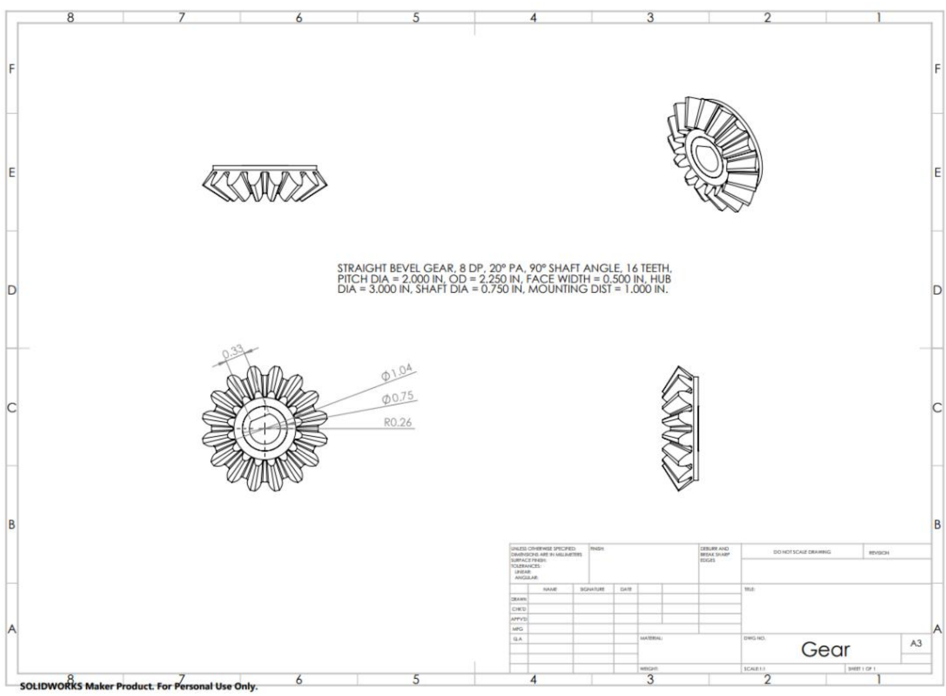

Figure A.12: Bevel pinion drawing with dimensions

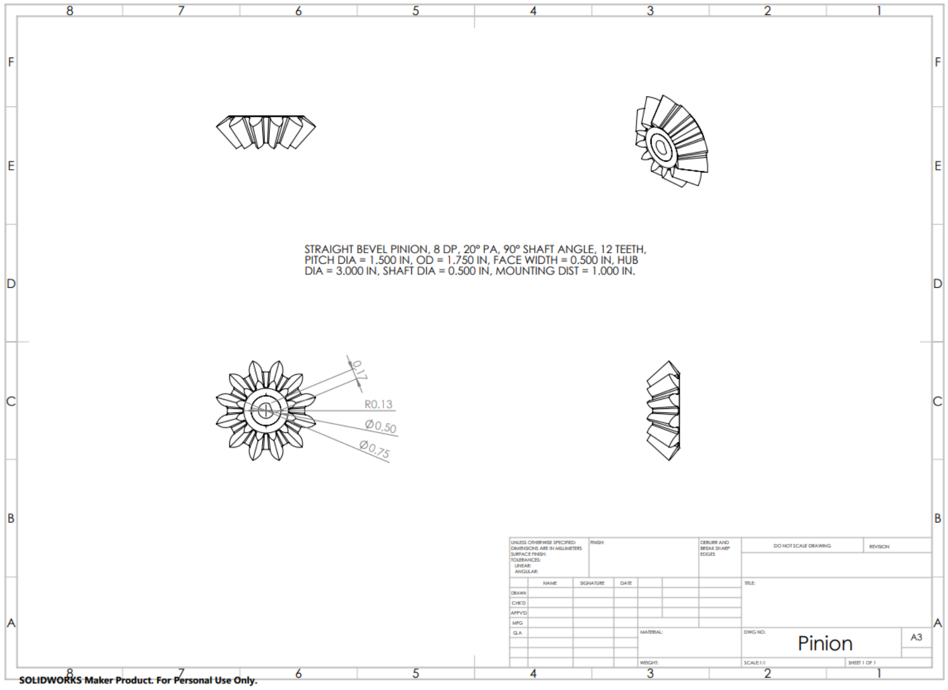

Figure A.13: Bevel gear drawing with dimensions

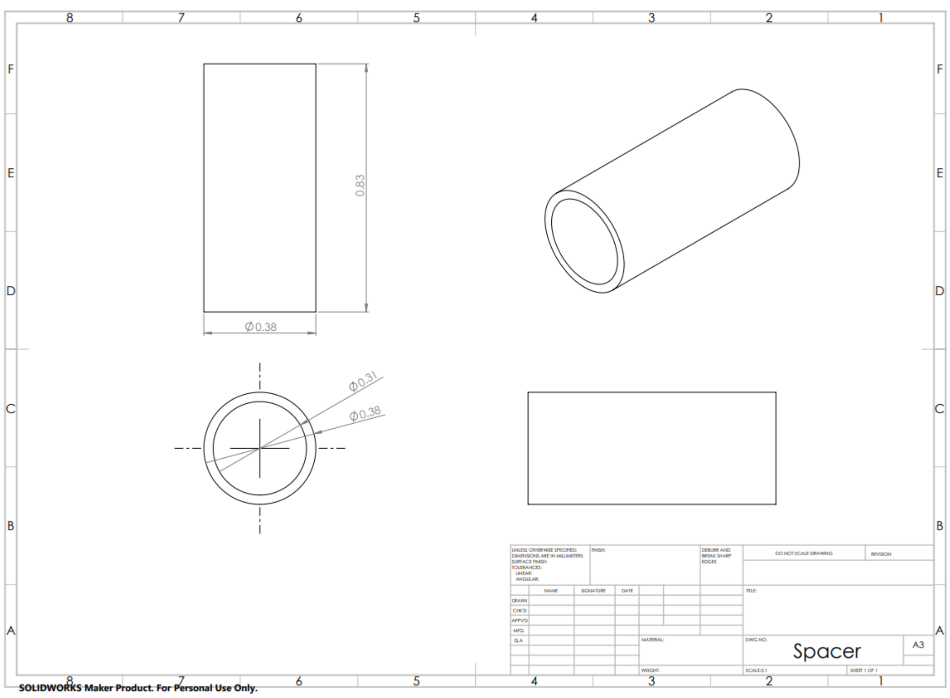

Figure A.14: Spacer drawing with dimensions